



COVER-KBC

Evidence-Centric Adaptive Inference for Closed-Book Knowledge Base Construction

Quoc-Linh Vo

Faculty of Information Technology, University of Science, VNU-HCM, Vietnam
vqlinh2433@apcs.fitus.edu.vn



Code & paper

Trust what's right. Keep asking what's missing.

Problem

- **Closed-book KBC:** for a query (s, r) , predict the full object set with one LLM ($\leq 32B$), no retrieval or fine-tuning.
- **AKBC 2026:** six relations whose answers may be empty, a single entity, a number, or a long list.
- **The hard part:** correct objects are not enough; a system must also judge if the set is complete.

Four relation types

SMALLSET

Need precise set closure
land borders, stock exchanges

NULLSINGLE

Empty answer vs. failed recall
city of death

NUMERIC

Cluster noisy numeric values
area, venue capacity

LARGEOPENSET

Keep discovering, avoid tail noise
award winners

Key idea

Voting over LLM samples shows what looks confident, not what is missing. COVER-KBC asks two separate questions:

1. **Is this answer correct?** Candidate confidence: support from different ways of asking, checked by a blind verifier.
2. **Is anything missing?** Residual search need: unstable sets, open existence checks or rival numbers keep the query open.

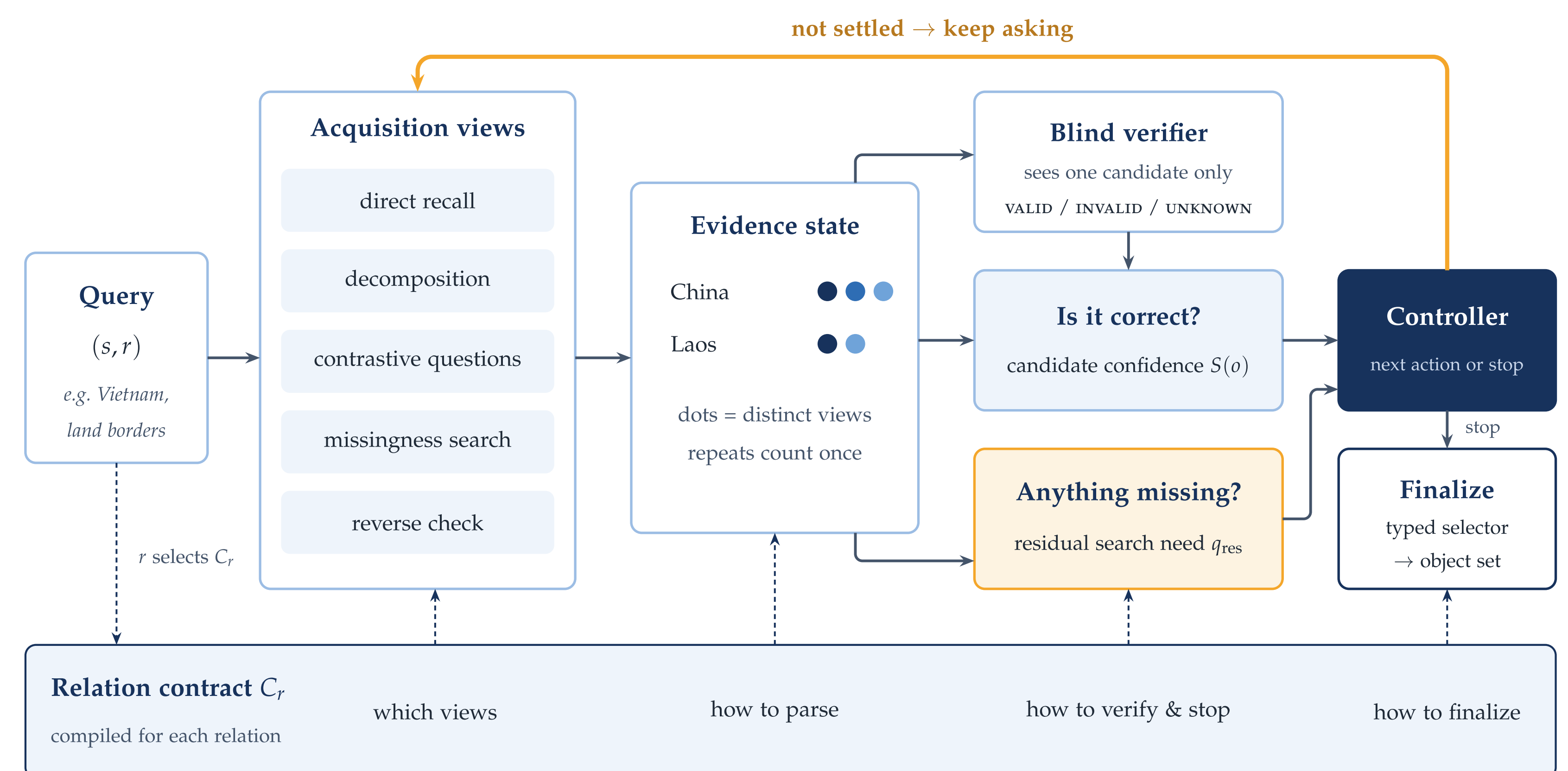
Example: for Vietnam's land borders, the LLM returns China and Laos. Both are correct, yet Cambodia is missing, so residual search need keeps the query open and another view is run.

What makes it work

- **Votes $\neq$ evidence:** repeats of one prompt count once.
- **Empty can be right:** for city of death, ask "Is the person deceased?" before asking "Where?".
- **Doubt counts against:** an UNKNOWN lowers the score.

Method

Each relation is compiled into an executable program. Every LLM call becomes evidence, and a budget-aware controller decides what to ask next and when to stop.



Is it correct? Distinct views, not repeats, build support; contradiction weighs 1.5x.

$$S(o) = \underbrace{\alpha F(o)}_{\text{independent support}} + \underbrace{\beta L(o)}_{\text{blind verifier}} - \underbrace{\delta C(o)}_{\text{contradiction}} - \underbrace{\eta U(o)}_{\text{template disagreement}}$$

$$F(o) = \frac{|g(o)|}{|m(o)|}, \quad (\alpha, \beta, \delta, \eta) = (1, 0.6, 1.5, 1)$$

Anything missing? One open gap is enough: a max over soft, mandatory and typed gaps.

$$q_{\text{res}} = \text{clip}_{[0,1]} \left(\max \{ \phi_w, \phi_{\text{mand}}, \phi_{\text{block}} \} \right)$$

$$\phi_{\text{block}} = \begin{cases} s(c_2) / s(c_1) & \text{NUMERIC: rival value cluster} \\ \max(u_{\exists}, \kappa_{\text{loc}}) & \text{NULLSINGLE, unless surely empty} \\ 0 & \text{otherwise} \end{cases}$$

Blind verifier: logits recalibrated on a content-free prompt; doubt counts against o.

$$\tilde{p}_j = \frac{\exp((z_j - b_j)/T)}{\sum_k \exp((z_k - b_k)/T)}$$

$$L(o) = \text{clip}_{[-1,1]} \left(\frac{1}{3} \log \frac{\tilde{p}_{\text{VALID}} + \epsilon}{\tilde{p}_{\text{INVALID}} + \tilde{p}_{\text{UNKNOWN}} + \epsilon} \right)$$

Controller: picks the best legal action under a 12-call budget, or stops.

$$a_t^* = \arg \max_{a \in \mathcal{A}_t} \underbrace{\lambda_y Y_t(a)}_{\text{yield}} + \underbrace{\lambda_g G_t(a)}_{\text{gap closure}} + \underbrace{\lambda_i I_t(a)}_{\text{info gain}} - \underbrace{\lambda_c \text{cost}(a)}_{\text{call cost}} - \underbrace{\lambda_d D_t(a)}_{\text{redundancy}}$$

Finalization: only evidence-backed candidates are output; numbers merge within 2.5%.

Results

Setup: one frozen Mistral-Small-3.2-24B checkpoint for every role (nf4, greedy decoding); no retrieval, no fine-tuning; official AKBC 2026 evaluation.

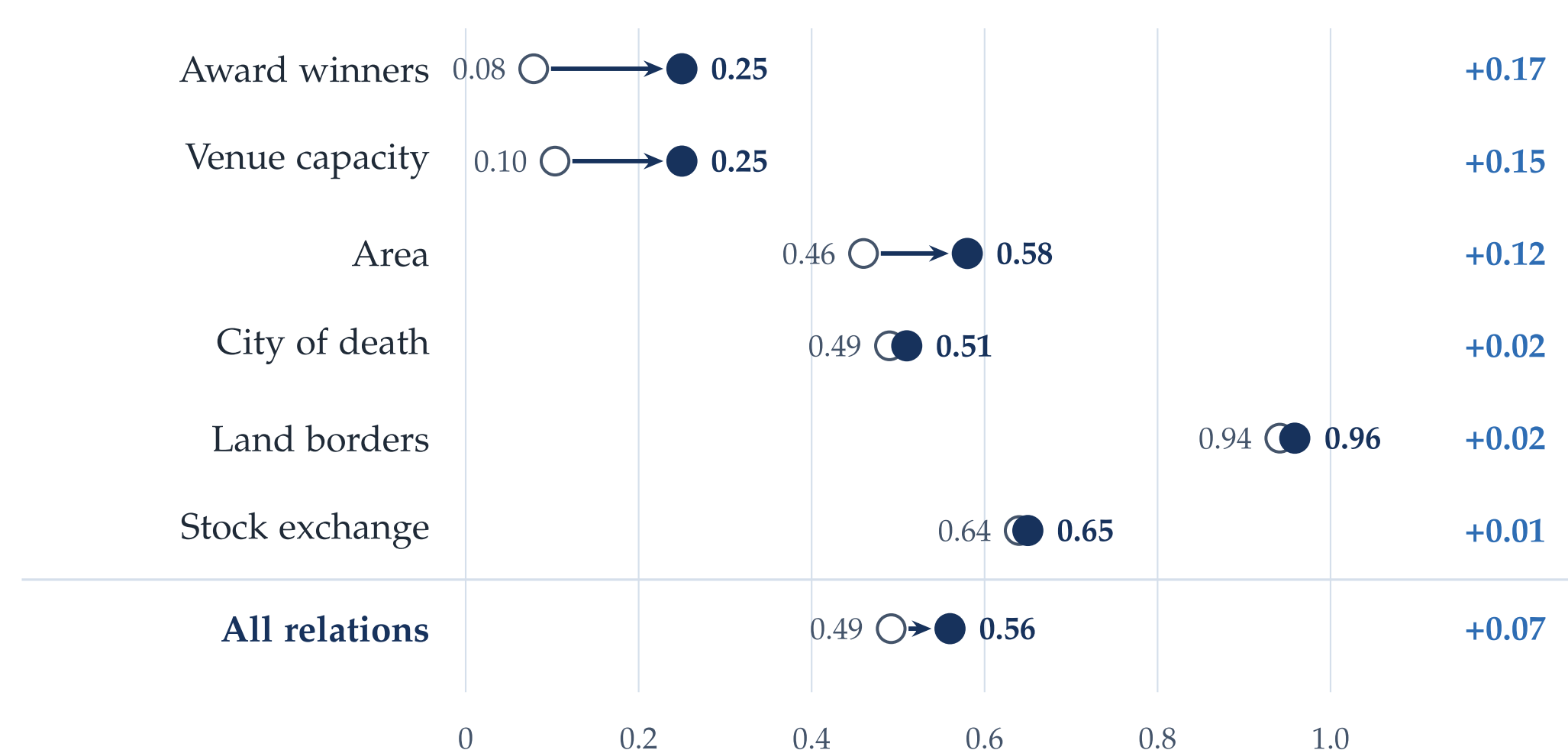
0.5878

Macro-F1 on the hidden test

Precision 0.7268, Recall 0.6055
+0.068 over direct prompting (validation)

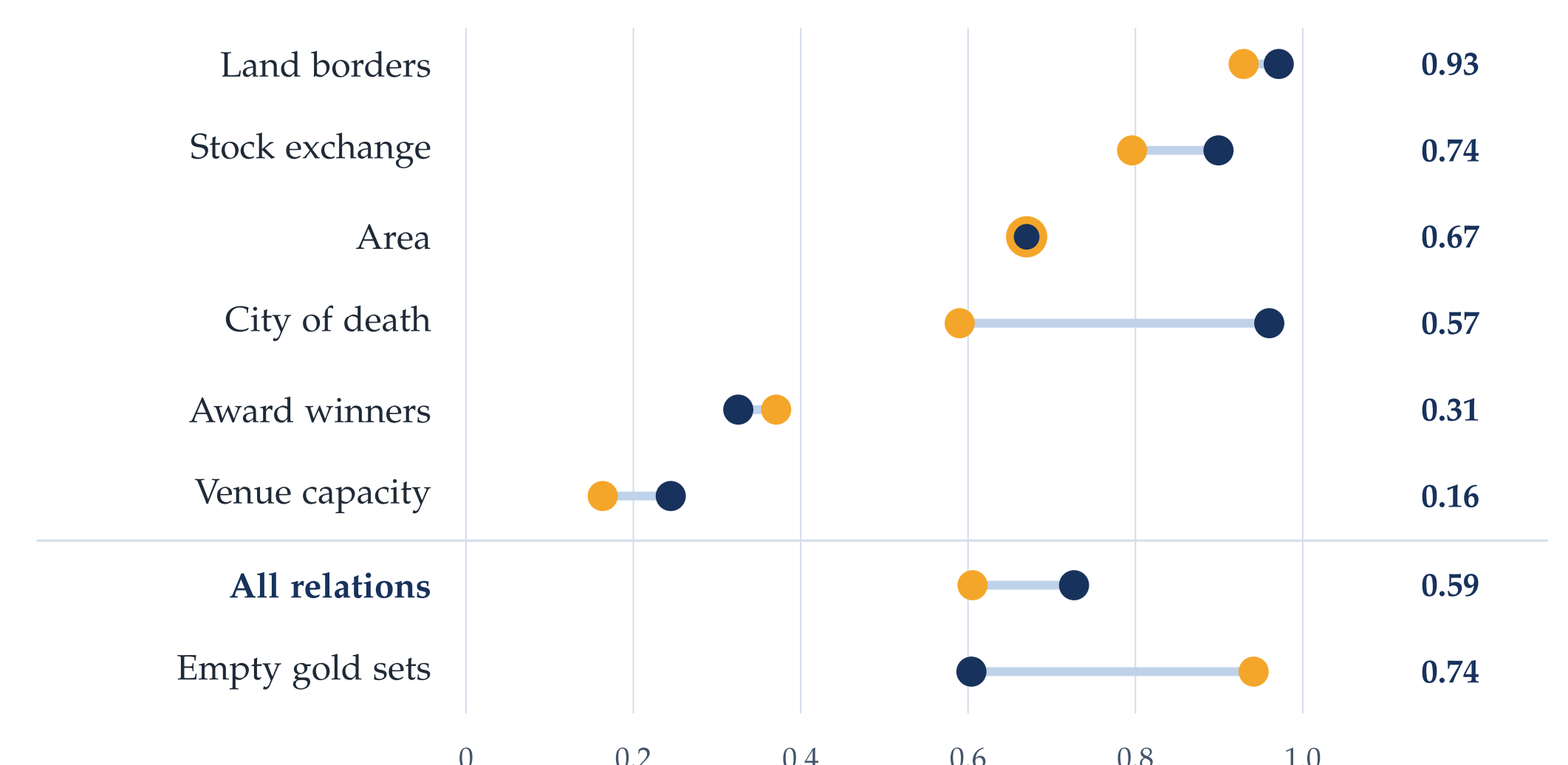
Same LLM, direct prompting vs. COVER-KBC

○ direct 5-shot ● COVER-KBC (validation Macro-F1)



Hidden test: precision and recall

● precision ● recall (per-relation macro averages)



Key findings

- **Closed small sets are strongest:** land borders reach 0.93 F1 with 0.97 precision.
- **Largest gains over direct prompting:** award winners +0.17, venue capacity +0.15, area +0.12.
- **Conservative by design:** precision beats recall on most relations, yet empty gold sets reach 0.94 recall.

Limitations

- **Venue capacity (0.16 F1):** the model rarely commits to the maximum published capacity.
- **Award winners (0.31 F1):** long recipient lists stay recall-limited without retrieval.
- **Cost:** contracts are hand-specified per relation; multi-step inference costs more than one prompt.

Acknowledgments: This research is partially supported by the Faculty of Information Technology, University of Science, VNU-HCM.